

Pedro Aquino da Costa

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EDUCATION

Universidade Federal Fluminense (UFF)

Niterói, Brazil

B.Sc. in Computer Science – in progress

- Coursework: Computer Architecture, Digital Circuits, Operating Systems, Arduino Kit Development

SENAI Taguatinga

Brasília, Brazil

Technical Degree in Electrotechnics

- Elected class vice-president; recognized as Model Student in every semester

AWARDS

1st Place – COPPE AI Hackathon 2026 (COPPE/UFRJ)

Sep. 2026

Developer – Team 20 | Atanora – Project Predicta

Rio de Janeiro, Brazil

- MVP that forecasts Brazilian grid demand from operator and weather data and turns it into a **dynamic tariff**
- Per-horizon models (H01–H24) for 4 grid subsystems with **Ridge** and **XGBoost**, validated against DESSEM
- Two engines (forecast and auditable tariff on regulated rates) linked by a data contract; **Python/Django**

EXPERIENCE

Fidelity Labs (Fidelitiesquisas)

Remote

Software Developer

- Led the migration of two legacy systems (Delphi 2010 and PHP/Docker) to a **Go** REST API with **PostgreSQL**, removing direct application authentication against MariaDB
- Moved file handling from FTP to an API across dozens of Delphi forms, with an XML client (Indy 10) and Go endpoints integrated with **DynamoDB**, **S3** and **AWS SSM**
- Solved critical cross-stack integration issues (JSON parsing in Delphi 2010, SSL in PHP/Docker containers); built onboarding tooling

DupTec

Remote – São Paulo, Brazil

Junior Developer

- Delphi/Pascal desktop applications, SQL queries and Python automation of data-processing routines

DriveIT

Remote – São Paulo, Brazil

Technical Analyst – technical support for customers and internal teams

PROJECTS

Chess AI (chess-ia) | Go, GitHub Actions

2026

- Custom engine with **bitboards**, **magic bitboards** and Zobrist hashing, validated with **perft** tests
- Alpha-beta with iterative deepening, quiescence search, transposition table and **PolyGlot** opening book
- HTTP worker with retry/backoff integrated with the game backend; CI with automated tests

RISC-V Pipeline Simulator | Python, RISC-V Assembly

2025

- **RV32I** simulator with a 5-stage pipeline, two-pass assembler (binary/hex output) and cycle-by-cycle log
- **Forwarding**, hazard stalls and a **2-bit branch predictor**: from ~45 to ~25 cycles on the test program

EXTRACURRICULAR ACTIVITIES

Blackbird Aerodesign – UFF Aerodesign Team

Niterói, Brazil

Member – Management & Performance Cell

- RC aircraft for SAE Brasil AeroDesign; moment of inertia testing with a bifilar pendulum

“Seja Digital” Program – Brazilian Federal Government

Volunteer digital inclusion instructor

TECHNICAL SKILLS

Languages: Go, C++, Python, Delphi/Pascal, JavaScript/TypeScript, PHP, SQL, RISC-V Assembly

Embedded and Data: Arduino, electronics; XGBoost, scikit-learn, pandas

Frameworks and Tools: Django, Flask, Next.js, Node.js, Raylib, Git, GitHub Actions, Docker

Databases and Cloud: PostgreSQL, MariaDB, DynamoDB, S3, AWS SSM

Languages Spoken: Portuguese (native), English (intermediate – reading and writing)